

Multiplication

A Simplify:

$$1 \quad \frac{1}{2} \times \frac{x}{3}$$

$$2 \quad \frac{1}{3} \times \frac{a}{2}$$

$$3 \quad \frac{x}{4} \times \frac{1}{3}$$

$$4 \quad \frac{a}{2} \times \frac{b}{3}$$

$$5 \quad \frac{x}{4} \times \frac{y}{5}$$

$$6 \quad \frac{a}{2} \times \frac{a}{3}$$

$$7 \quad \frac{a}{4} \times \frac{c}{4}$$

$$8 \quad \frac{a}{2} \times \frac{4}{a}$$

$$9 \quad \frac{a}{3} \times \frac{3}{a}$$

$$10 \quad \frac{c}{2} \times \frac{4}{c}$$

B Simplify:

$$1 \quad \frac{2a}{3} \times \frac{3}{2}$$

$$2 \quad \frac{p}{q} \times \frac{a}{b}$$

$$3 \quad \frac{a}{5} \times \frac{10}{c}$$

$$4 \quad \frac{xy}{3} \times \frac{2}{x}$$

$$5 \quad \frac{ab}{4} \times \frac{2}{a}$$

$$6 \quad \frac{a}{3} \times \frac{3}{c} \times \frac{c}{a}$$

$$7 \quad \frac{5a}{6} \times \frac{3}{10}$$

$$8 \quad \frac{3}{4} \times \frac{8a}{7}$$

$$9 \quad \frac{5c}{6} \times \frac{3}{10c}$$

$$10 \quad \frac{9y}{8} \times \frac{4}{3x}$$

$$11 \quad 7 \times \frac{3x}{14}$$

$$12 \quad \frac{6x}{y} \times \frac{2y}{15}$$

$$13 \quad \frac{7x}{8} \times \frac{3y}{14x}$$

$$14 \quad \frac{9c}{4} \times \frac{8c}{3}$$

$$15 \quad \frac{11a}{12} \times \frac{6}{a} \times \frac{4c}{11}$$

$$16 \quad \frac{3x^2}{5} \times \frac{y}{x}$$

$$17 \quad \frac{2x}{y^2} \times \frac{y}{x}$$

$$18 \quad 6x^2 \times \frac{1}{4x^2}$$

$$19 \quad \left(\frac{2x}{3}\right)^2$$

$$20 \quad \left(\frac{3x}{4}\right)^2$$

$$21 \quad \left(\frac{a}{2}\right)^3$$

$$22 \quad \left(\frac{a}{2}\right)^2 \times \frac{4}{a}$$

$$23 \quad \left(\frac{x}{3}\right)^2 \times \frac{9}{x^2}$$

$$24 \quad \frac{x}{6} \times \frac{3}{x} \times \frac{y}{4}$$

$$25 \quad \frac{a+1}{2} \times \frac{4}{3}$$

$$26 \quad \frac{x-1}{2} \times \frac{4}{x-1}$$

$$27 \quad \frac{1}{3}(a-3) \times 12$$

$$28 \quad \frac{a+2}{6} \times \frac{12}{a+2}$$

$$29 \quad \frac{9x}{8} \times \frac{4(x-1)}{3}$$

$$30 \quad \frac{3}{2a} \times \frac{2c}{3} \times \frac{a}{c}$$

C Simplify:

$$1 \quad \frac{x-3}{x} \times \frac{x}{x+3}$$

$$2 \quad \frac{a}{a+1} \times \frac{a+1}{a^2}$$

$$3 \quad \frac{c^2}{c-2} \times \frac{c-2}{c}$$

$$4 \quad \frac{ac}{c+1} \times \frac{a+1}{ac}$$

$$5 \quad \frac{x}{x+y} \times \frac{x+y}{x}$$

$$6 \quad \frac{y^2}{x-y} \times \frac{x-y}{y}$$

$$7 \quad \frac{a}{a-b} \times \frac{2a-2b}{a}$$

$$8 \quad \frac{4x^2}{x+1} \times \frac{x+1}{2x}$$

$$9 \quad \frac{a^2+a}{c} \times \frac{c}{a+1}$$

$$10 \quad \frac{a^2-1}{3} \times \frac{6}{a-1}$$

$$11 \quad \frac{x^2-4}{5} \times \frac{3}{x-2}$$

$$12 \quad \frac{x+2}{4} \times \frac{8}{x^2-4}$$

$$13 \quad \frac{9a}{a^2-1} \times \frac{a+1}{3a^2}$$

$$14 \quad \frac{2m}{m^2+2m+1} \times \frac{m+1}{4m}$$

$$15 \quad \frac{x^2-x-6}{3} \times \frac{6}{x-3}$$

$$16 \quad \frac{a^2-1}{a+1} \times \frac{3}{a-1}$$

Division

A Simplify:

$$1 \quad \frac{a}{2} \div \frac{a}{4}$$

$$2 \quad \frac{a}{4} \div \frac{a}{2}$$

$$3 \quad \frac{x}{3} \div \frac{x}{2}$$

$$4 \quad \frac{x}{4} \div \frac{x}{3}$$

$$5 \quad \frac{2}{c} \div \frac{4}{c}$$

$$6 \quad \frac{3}{y} \div \frac{1}{y}$$

$$7 \quad \frac{5}{c} \div \frac{3}{c}$$

$$8 \quad 2 \div \frac{2}{x}$$

$$9 \quad \frac{x}{3} \div 2$$

$$10 \quad \frac{2a}{3} \div \frac{4}{9}$$

B Simplify:

$$1 \quad \frac{5x}{3} \div \frac{2x}{9}$$

$$2 \quad \frac{3a}{4} \div \frac{a}{8}$$

$$3 \quad \frac{5c}{6} \div \frac{2c}{3}$$

$$4 \quad a \div \frac{2a}{3}$$

$$5 \quad \frac{x}{y} \div \frac{2x}{y}$$

$$6 \quad \frac{5d}{3} \div \frac{10}{3}$$

$$7 \quad \frac{3}{4} \div \frac{7a}{8}$$

$$8 \quad \frac{3}{10c} \div \frac{6}{5c}$$

$$9 \quad \frac{8}{9y} \div \frac{4}{3x}$$

$$10 \quad \frac{1}{7} \div \frac{5x}{14}$$

$$11 \quad \frac{y}{6x} \div \frac{2y}{15x}$$

$$12 \quad \frac{8}{7y} \div \frac{3x}{14y}$$

$$13 \quad \frac{4a}{9c} \div \frac{8a}{3}$$

$$14 \quad \frac{a^2b}{3} \div \frac{ab}{6}$$

$$15 \quad \left(\frac{a}{2}\right)^2 \div \frac{a}{8}$$

$$16 \quad \frac{3x^2}{5} \div \frac{6x}{y}$$

$$17 \quad \frac{y^2}{2x} \div \frac{y}{x}$$

$$18 \quad \frac{1}{6x^2} \div \frac{1}{9x^2}$$

$$19 \quad \frac{x}{6} \times \frac{3}{x} \div \frac{1}{2}$$

$$20 \quad \frac{2a}{3} \times \frac{1}{4a} \div \frac{1}{2a}$$

$$21 \quad \left(\frac{a}{4}\right)^2 \div \frac{3a}{8}$$

$$22 \quad \frac{3}{2c} \div \frac{7}{8c^2}$$

$$23 \quad \frac{x+1}{2} \div \frac{3}{4}$$

$$24 \quad \frac{x-1}{4} \div \frac{1}{8}$$

$$25 \quad \frac{3}{a+1} \div \frac{6}{a+1}$$

$$26 \quad \frac{2}{3(a+1)} \div \frac{4}{a+1}$$

$$27 \quad 7a^2b \div \frac{ab^2}{3}$$

$$28 \quad 9(a-1) \div 3(a-1)^2$$

$$29 \quad \frac{1}{3}(a-1) \div \frac{2}{3}a$$

$$30 \quad 4a^2 + 3a \div \frac{1}{2a}$$

C Simplify:

$$1 \quad \frac{a-1}{a} \div \frac{a+1}{a}$$

$$2 \quad \frac{a}{a+1} \div \frac{a^2}{a+1}$$

$$3 \quad \frac{a^2b}{2c} \div \frac{4ab^2}{c^2}$$

$$4 \quad \frac{a+1}{2} \div \frac{a+1}{3}$$

$$5 \quad \frac{a^2-1}{2} \div \frac{a+1}{4}$$

$$6 \quad \frac{3}{x+1} \div \frac{6}{x^2-1}$$

$$7 \quad \frac{a}{a^2-1} \div \frac{a^2}{a+1}$$

$$8 \quad \frac{4x^2}{x+1} \div \frac{2x}{x+1}$$

$$9 \quad \frac{a^2+a}{4} \div \frac{a+1}{8}$$

$$10 \quad \frac{y^2-4}{3} \div \frac{y+2}{6}$$

$$11 \quad \frac{d^2-9}{3} \div \frac{d-3}{9}$$

$$12 \quad \frac{x^2+2x+1}{4} \div \frac{x+1}{2}$$

$$13 \quad \frac{y^2+2y-8}{y} \div \frac{y+4}{y^2}$$

$$14 \quad \frac{c^2+6c+9}{3c} \div \frac{c^2-9}{4c}$$

$$15 \quad \frac{4a^2b^3}{3c^2} \div \frac{8ab^4}{5c}$$

$$16 \quad \frac{2d}{4d-1} \div \frac{3d^2}{8d-2}$$

18. Algebraic Fractions: Multiplication

A 1 $\frac{x}{6}$ 2 $\frac{a}{6}$ 3 $\frac{x}{12}$ 4 $\frac{ab}{6}$ 5 $\frac{xy}{20}$ 6 $\frac{a^2}{6}$ 7 $\frac{ac}{16}$

8 2 9 1 10 2

B 1 a 2 $\frac{ap}{bq}$ 3 $\frac{2a}{c}$ 4 $\frac{2y}{3}$ 5 $\frac{b}{2}$ 6 1 7 $\frac{a}{4}$
8 $\frac{6a}{7}$ 9 $\frac{1}{4}$ 10 $\frac{3y}{2x}$ 11 $\frac{3x}{2}$ 12 $\frac{4x}{5}$ 13 $\frac{3y}{16}$
14 $6c^2$ 15 $2c$ 16 $\frac{3xy}{5}$ 17 $\frac{2}{y}$ 18 $\frac{3}{2}$ 19 $\frac{4x^2}{9}$
20 $\frac{9x^2}{16}$ 21 $\frac{a^3}{8}$ 22 a 23 1 24 $\frac{y}{8}$ 25 $\frac{2(a+1)}{3}$

26 2 27 $4(a-3)$ 28 2 29 $\frac{3x(x-1)}{2}$ 30 1

C 1 $\frac{x-3}{x+3}$ 2 $\frac{1}{a}$ 3 c 4 $\frac{a+1}{c+1}$ 5 1 5 y 7 2

8 $2x$ 9 a 10 $2(a+1)$ 11 $\frac{3(x+2)}{5}$ 12 $\frac{2}{x-2}$

13 $\frac{3}{a(a-1)}$ 14 $\frac{1}{2(m+1)}$ 15 $2(x+2)$ 16 3

19. Algebraic Fractions: Division

A 1 2 $\frac{2}{3}$ 2 $\frac{1}{2}$ 3 $\frac{2}{3}$ 4 $\frac{3}{4}$ 5 $\frac{1}{2}$ 6 3 7 $\frac{5}{3}$ 8 x 9 $\frac{x}{6}$
10 $\frac{3a}{2}$

B 1 $\frac{15}{2}$ 2 6 3 $\frac{5}{4}$ 4 $\frac{3}{2}$ 5 $\frac{1}{2}$ 6 $\frac{d}{2}$ 7 $\frac{6}{7a}$ 8 $\frac{1}{4}$

9 $\frac{2x}{3y}$ 10 $\frac{2}{5x}$ 11 $\frac{5}{4}$ 12 $\frac{16}{3x}$ 13 $\frac{1}{6c}$ 14 $2a$

15 $2a$ 16 $\frac{xy}{10}$ 17 $\frac{y}{2}$ 18 $\frac{3}{2}$ 19 1 20 $\frac{a}{3}$ 21 $\frac{a}{6}$

22 $\frac{12c}{7}$ 23 $\frac{2(x+1)}{3}$ 24 $2(x-1)$ 25 $\frac{1}{2}$ 26 $\frac{1}{6}$

27 $\frac{21a}{b}$ 28 $\frac{3}{a-1}$ 29 $\frac{a-1}{2a}$ 30 $10a^2$

C 1 $\frac{a-1}{a+1}$ 2 $\frac{1}{a}$ 3 $\frac{ac}{8b}$ 4 $\frac{3}{2}$ 5 $2(a-1)$ 6 $\frac{x-1}{2}$

7 $\frac{1}{a(a-1)}$ 8 $2x$ 9 $2a$ 10 $2(y-2)$ 11 $3(d+3)$

12 $\frac{x+1}{2}$ 13 $y(y-2)$ 14 $\frac{4(c+3)}{3(c-3)}$ 15 $\frac{5a}{6bc}$

16 $\frac{4}{3d}$

20. Algebraic Fractions — Review

A 1 $\frac{6x}{5}$ 2 $\frac{2x}{3}$ 3 $\frac{a}{4}$ 4 $\frac{5a}{2}$ 5 $\frac{7a}{12}$ 6 1 7 $\frac{11x}{12}$

8 $\frac{2y}{3}$ 9 $\frac{7a}{12}$ 10 $\frac{8}{5}$ 11 3 12 $\frac{5a}{42}$ 13 $\frac{1}{6}$ 14 $\frac{a}{12}$

15 c^3 16 $\frac{4a^2}{25}$ 17 $\frac{3}{2a}$ 18 $\frac{x}{2}$ 19 $6c$ 20 $\frac{a-5}{2}$

21 $a+3$

B 1 $\frac{y}{2}$ 2 $\frac{x-y}{2}$ 3 $\frac{1}{3+a}$ 4 $\frac{4b}{5a}$ 5 $\frac{5a}{2}$ 6 $\frac{y}{3}$

7 $\frac{1}{2y-2}$ 8 $\frac{19d}{24}$ 9 $\frac{1}{ab^2}$ 10 $\frac{a+4}{2}$ 11 $\frac{18x}{35}$

12 $\frac{1}{8}$ 13 $\frac{10-c}{4}$ 14 $\frac{4a^2}{9}$ 15 2 16 $\frac{3a}{4}$ 17 6

18 $\frac{22y}{30}$ 19 2 20 $\frac{1}{4}$ 21 $\frac{5}{6a}$ 22 $\frac{5}{4}$ 23 $\frac{5}{6}$

24 $\frac{2(a-1)}{5}$ 25 $\frac{2a-b}{8}$ 26 $\frac{5}{6x}$ 27 $\frac{2a}{b}$ 28 $\frac{2}{3}$

29 $3x^2$ 30 $\frac{43}{30a}$

C 1 $\frac{8x+3}{6}$ 2 $\frac{2x}{3}$ 3 a 4 $\frac{7x-10}{12}$ 5 y 6 $2a$

7 $\frac{17x+5}{20}$ 8 $\frac{x-2}{6}$ 9 $3ab$ 10 $\frac{17x-3}{21}$ 11 $\frac{16xy}{5}$

12 $\frac{a}{2}$ 13 $x+1$ 14 $\frac{13}{6a}$ 15 $\frac{x+1}{2}$ 16 $\frac{a+19}{12}$

17 $\frac{a^2b}{2}$ 18 $\frac{5x-1}{6}$ 19 $\frac{3b}{4}$ 20 $\frac{a+3}{2(a-3)}$ 21 2

22 $2(x+4)$ 23 $\frac{1}{a-2}$ 24 $\frac{a-7}{20}$

21. Simultaneous Equations: Substitution

A 1 $x=2, y=4$ 2 $x=2, y=6$

B 1 $x=3, y=3$ 2 $x=2, y=6$ 3 $x=-2, y=-8$

4 $x=4, y=-4$ 5 $x=2, y=4$ 6 $x=7, y=-14$

7 $x=2, y=3$ 8 $x=2, y=0$ 9 $x=3, y=12$

10 $x=3, y=9$ 11 $x=-8, y=-4$ 12 $x=2, y=6$

13 $x=3, y=6$ 14 $x=3, y=2$

C 1 $x=-4, y=2$ 2 $x=3, y=0$ 3 $x=2, y=-6$

4 $x=6, y=-9$ 5 $x=-2, y=3$ 6 $x=2, y=0$

7 $x=2, y=4$ 8 $x=-2, y=-1$

22. Simultaneous Equations: Elimination

A 1 $x=2, y=-1$ 2 $x=-2, y=5$

B 1 $x=3, y=1$ 2 $x=2, y=0$ 3 $x=1, y=0$

4 $x=1, y=1$ 5 $x=-2, y=-1$ 6 $x=-1, y=4$

7 $x=-2, y=5\frac{1}{2}$ 8 $x=\frac{5}{3}, y=1$ 9 $a=-2, b=-1$

10 $x=-5, y=4\frac{1}{2}$

C 1 C 2 $x=-2, y=2$ 3 $a=12\frac{1}{2}, b=6$

4 $x=5, y=-4\frac{1}{2}$ 5 $x=-1, y=-1$ 6 $x=2, y=-2\frac{1}{2}$

7 $x=0, y=-3$ 8 $x=3, y=-5$ 9 $x=1, y=-8$

23. Elimination by Matching Co-efficients

A 1 $x=6, y=-7$ 2 $x=-1, y=2$

B 1 $x=2, y=1$ 2 $x=2, y=2$ 3 $x=1, y=2$

4 $a=-2, b=3$ 5 $x=-1, y=4$ 6 $x=5, y=4$

7 $x=-3, y=-2$ 8 $c=-1, d=-2$ 9 $x=1, y=0$

10 $x=2, y=-4$ 11 $a=1, b=3$ 12 $x=1, y=1$

13 $x=1, y=-1$ 14 $x=2, y=1$

C 1 $x=2, y=2$ 2 $x=-7, y=-3$ 3 $x=-2, y=-5$

4 $a=21, b=17$ 5 $a=-3, b=1$ 6 $x=1, y=2$

24. One Linear and One Non-linear

A 1 $y=4, 1$ 2 $x=-3, 2; y=9, 4$

B 1 (1, 1), (-2, 4) 2 (3, 9), (-2, 4) 3 (4, 16), (-1, 1)

4 (4, 16), (-2, 4) 5 (2, 4), (1, 1) 6 (5, 25), (1, 1)

7 (6, 36), (1, 1) 8 (4, 16), (-3, 9) 9 (3, 9), (-4, 16)

10 (8, 64), (-2, 4) 11 (-1, 1) and (-4, 16)

12 (2, 4) and (-5, 25) 13 (-4, 16) and (5, 25)

14 (-6, 36) and (4, 16) 15 (-1, 1) and (3, 9)

16 (2, 4) and (-9, 81)

C 1 (5, 2), (-2, -5) 2 (2, 1), (-1, -2) 3 (0, 5), (5, 0)

4 (1, 3), $(-\frac{9}{5}, -\frac{13}{5})$ 5 (2, 3), (3, 2) 6 (0, -2), (3, 4)

25. Straight Line Graphs

A A(3, 2), B(2, 4), C(-2, 2), D(-3, 1), E(2, -1), F(1, -3),
G(1, 0), H(0, 3), J(-2, -1), K(0, -2)

1 2, 4, 6 2 -3, -1, 1 3 -4, 0, 6 4 -1, 0, 2

5 3, 7, 9 6 -2, -4, -6

B 13 2, 5 14 3, -2 15 4, -3 16 -3, 5 17 1, 1

18 2, 0 19 -1, 0 20 -3, -4 21 $\frac{1}{2}, 3$ 22 $\frac{1}{3}, -1$

23 $-\frac{1}{4}, 2$ 24 $-\frac{1}{2}, -5$ 25 $y=2x+3$ 26 $y=3x-2$

27 $y=-x-1$ 28 $y=-2x+4$ 29 $y=\frac{1}{3}x+1$

30 $y=-\frac{1}{2}x-2$

C 1 $y=-2x+1, -2, 1$ 2 $y=3x-2, 3, -2$

3 $y=-5x+7, -5, 7$ 4 $y=x-4, 1, -4$

5 $y=4x+5, 4, 5$ 6 $y=-\frac{1}{2}x+1, -\frac{1}{2}, 1$